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<https://github.com/maitelle-cell/PDAN8411>

ICE TASK 4: Classification and Model Improvement

1. Dataset Justification

A suitable dataset is the UBC Ovarian Cancer Subtype Classification and Outlier Detection (UBC-OCEAN) dataset available through Kaggle (<https://www.kaggle.com/code/jocelyndumlao/ovarian-cancer-classification-with-cnns#-Dataset-Overview>). The dataset contains ovarian cancer images that can be used to classify different ovarian cancer subtypes. The Kaggle notebook shows exploratory data analysis and the use of a Convolutional Neural Network (CNN) for image classification.

This dataset is suitable as the target is a classification problem: the model learns from labelled medical images and predicts the relevant cancer class/subtype.

2. Algorithm Choice

I would use a Convolutional Neural Network (CNN) because CNNs are designed for image classification. They can automatically learn important visual features such as shapes, textures and patterns from medical images.

CNN based approaches have been explored for ovarian cancer classification using histopathological images.

3. Analysis Plan

Before training the model, I would take the following three steps:

1. Exploratory Data Analysis (EDA):

I will look at the number of images in each cancer class, look at sample images, check if dataset balanced or not.

2. Data cleaning and preprocessing:

Remove unsuitable/corrupted images, resize images to a consistent size, normalise pixel values. The data would then be split into training, validation and testing sets.

3. Feature preparation and selection:

For a CNN, image features would be automatically learned. I would also look at the class distribution and use data augmentation if some classes had fewer images.

4. Model Evaluation

I would use Recall and F1-score.

- Recall: Important in cancer classification as it measures how many actual cancer cases are correctly identified. A low recall would mean cancer cases are being missed.
- F1-score: Combines precision and recall, is useful when the classes are imbalanced. Research on ovarian cancer classification also uses recall and F1-score along with other classification metrics.

5. Model Improvement

If the model did not perform well, I would use data augmentation and hyperparameter tuning. Data augmentation would add more variety of training images, tuning the learning rate/batch size/CNN layers may improve the classification performance.

Conclusion

Overall, the UBC-OCEAN ovarian cancer dataset provides a suitable classification problem since labelled medical images can be used to train a CNN to identify cancer categories. The workflow would involve EDA → preprocessing → model training → evaluation → model improvement.

Reference

- Jocelyn Dumlao. *Ovarian Cancer Classification with CNNs*. Kaggle, based on the UBC Ovarian Cancer Subtype Classification and Outlier Detection (UBC-OCEAN) competition

(<https://www.kaggle.com/code/jocelyndumlao/ovarian-cancer-classification-with-cnns#-Dataset-Overview>).

- Delen, D., & Demirkan, H. (2020). *Big data analytics methods: A 2nd edition*. Springer.